

1. If $g_2(t) = k g_1(t)$, the energy of the signal $g_2(t)$ equals:

a) The energy of $g_1(t)$

☒ b) The energy of $g_1(t)$ multiplied by k^2

b) The energy of $g_1(t)$ multiplied by $2k$

d) The energy of $g_1(t)$ multiplied by k

2. The unit impulse response of a linear time invariant system is the unit step function $u(t)$. for $t > 0$, the response of the system to the input $e^{-at}u(t)$, $a > 0$, will be

(A) ae^{-at}

☒ (B) $\frac{1-e^{-at}}{a}$

(C) $a(1 - e^{-at})$

(D) $1 - e^{-at}$

3. the continuous time system described by $y(t) = x(t^2)$ is

☒ (A) Causal, linear and time varying.

☒ (B) Causal, non-linear and time varying.

☒ (C) Non-causal, non-linear and time invariant.

☒ (D) Non-causal, linear and time invariant.

4. Given a unit step function $u(t)$, its time derivative is:

☒ (A) Unit impulse.

(B) Another step function

☒ (C) A unit ramp function

(D) A sine function

5. the area under the curve $\int_{-\infty}^{\infty} \delta(t) dt$ is

(A) ∞

☒ (B) Unity

☒ (C) 0

(D) Undefined

6. An LTI system has impulse response $h(t) = u(t-1)$, If the input is $x(t) = u(t-2)$ the output $y(t)$ is given by

(A) $y(t) = tu(t)$

(B) $y(t) = tu(t-1)$

☒ (C) $y(t) = (t-1)u(t-1)$

☒ (D) $y(t) = (t-3)u(t-3)$

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7. Consider a series interconnection of two LTI systems that have the impulse Responses $h_1[n] = \delta[n]$ and $h_2[n] = \delta[n] - \delta[n-2]$, if the input is $x[n] = u[n]$. What is the output of the combined system?

☒ (A) $y[n] = \delta[n]$

☒ (B) $y[n] = \delta[n] + \delta[n-1]$

☒ (C) $y[n] = \delta[n-1] + \delta[n-2]$

☒ (D) $y[n] = \delta[n] + \delta[n-1] + \delta[n-2]$

8. the fundamental period and frequency of the following signal $x(t) = 2 \cos(10\pi t + \frac{\pi}{6})$

(a) $T_0 = 0.1 \text{ sec}, f_0 = 10 \text{ Hz}$

(b) $T_0 = 10 \text{ sec}, f_0 = 0.1 \text{ Hz}$

☒ (c) $T_0 = 0.2 \text{ sec}, f_0 = 5 \text{ Hz}$

(d) $T_0 = 5 \text{ sec}, f_0 = 0.2 \text{ Hz}$

10. The $x(\alpha t)$ is a compressed version of $x(t)$ if:

A. $|\alpha| < 1$

☒ B. $|\alpha| > 1$

C. $|\alpha| \neq 1$

D. $|\alpha| = 1$

11. If $x(t)$ has fundamental period $T = 10$, then $x(\frac{1}{2}t - 8)$ has fundamental period

A. 5

B. 20

C. 10

D. It is not periodic

12. The output of the discrete time system $h[n] = \{1 \ 3 \ 2 \ 2\}$; when the input $x[n] = \{1 \ -2 \ 3\}$ is applied

(A) $y[n] = \{1 \ -1 \ 7 \ 2 \ 6\}$

(B) $y[n] = \{1 \ 1 \ 7 \ 4 \ 6\}$

(C) $y[n] = \{1 \ 1 \ -1 \ 7 \ -4 \ 6\}$

(D) $y[n] = \{1 \ 1 \ 1 \ 7 \ 2 \ 6\}$

13. A system having impulse response $h(t)$ will be BIBO stable if

(A) $\int_{-\infty}^{\infty} |h(t)| dt > \infty$

(B) $\int_{-\infty}^{\infty} h(t) dt < -\infty$

(C) $\int_{-\infty}^{\infty} |h(t)| dt < \infty$

(D) $\int_{-\infty}^{\infty} |h(t)| dt = \infty$

14. In these systems any time delay in input signal leads to identical time delay in the output signal :

A. Time invariant systems	B. Memory less systems
C. Causal systems	D. Non-causal systems

15. Which system is defined as causal system of the following:

(A) $y(n) = \frac{1}{3}x(n) + x(n-1) + x(n+1)$

(B) $y(n) = \frac{1}{3}x(n) + x(n-1) + x[n - (n-2)]$

16. Find the correct representation for the following:

(A) $x_1(n) * x_2(n) = x_2(n) * x_1(n)$

(B) $x_1(n) * [x_2(n) * x_3(n)] = [x_1(n) * x_2(n)] + x_3(n)$

(C) $x_1(n) * [x_2(n) + x_3(n)] = [x_1(n) * x_2(n) + x_1(n)] + x_3(n)$

(D) all are valid

17. Evaluate the integration value $\int_{-\infty}^{\infty} \sin(3(t-1))\delta(t)dt$

(A) $\frac{-1}{3} \sin 7$

b) $\frac{1}{3} \cos 5$

(C) $\frac{1}{6} \sin 7$

d) $\frac{-1}{6} \cos 7$

e) $-\sin 3$

18. The value of integration $\int_{-\infty}^{\infty} e^{2-t} \cdot \delta(t-2)dt$

a) **2**

b) 5

c) 1

d) 0

19. Which one is discrete time periodic signal?

a- $\sin \sqrt{3}\pi n$

b- $\cos \sqrt{2}\pi n$

c- $\sin 3\pi n$

d- All of them are correct

20. $y(t) = x(8t)$ is

a) Compressed signal

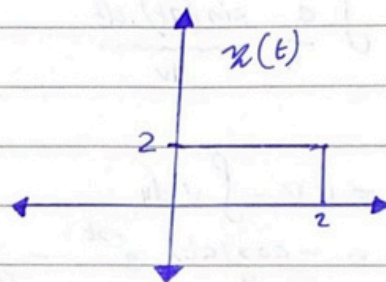
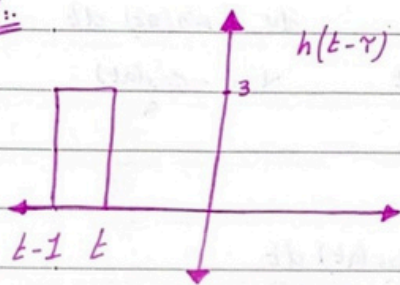
b) Expanded signal

c) Shifted signal

d) Amplitude scaled signal by a factor of 2

2] Find The impulse response of the following system:

Sol:



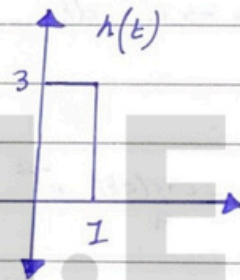
$$t < 0 \Rightarrow y(t) = 0$$

$$t > 0 \text{ \&ff; } t < 1 \Rightarrow y(t) = \int_0^t 6 \cdot d\tau = 6t$$

$$t > 1 \text{ \&ff; } t < 2 \Rightarrow y(t) = \int_{t-1}^t 6 \cdot d\tau = 6$$

$$t > 2 \text{ \&ff; } t < 3 \Rightarrow y(t) = \int_{t-1}^t 6 \cdot d\tau = 18 - 6t$$

$$t > 3 \Rightarrow y(t) = 0$$



$$y(t) = \begin{cases} 0, & t < 0 \\ 6t, & 0 \leq t < 1 \\ 6, & 1 \leq t < 2 \\ 18-6t, & 2 \leq t < 3 \\ 0, & t \geq 3 \end{cases}$$